
Module 1

Introduction to Computers

1.1 Computer

- ✓ A computer can be defined as an electronic device that is designed to accept data, perform the required mathematical and logical operations at high speed, and output the result.

1.2 Characteristics of Computers

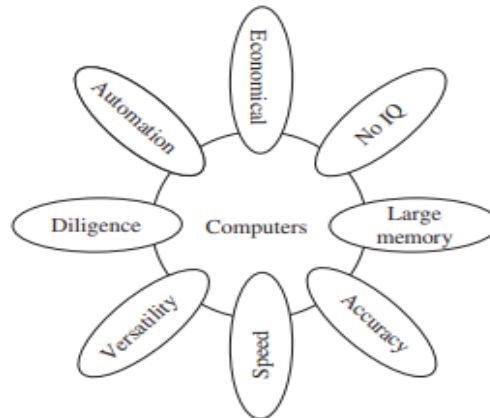


Figure 1.1. Characteristics of computers

1. Speed

- ✓ Computers can perform millions of operations per second.
- ✓ The speed of computers is usually given in nanoseconds and picoseconds, where 1 nanosecond = 1×10^{-9} seconds and 1 picosecond = 1×10^{-12} seconds.

2. Accuracy

- ✓ A computer is a very fast, reliable, and robust electronic device.
- ✓ It always gives accurate results, provided the correct data and set of instructions are input to it.
- ✓ This clearly means that the output generated by a computer depends on the given instructions and input data.
- ✓ If the input data is wrong, then the output will also be erroneous. In computer terminology, this is known as garbage-in, garbage-out (GIGO).

3. Automation

- ✓ Besides being very fast and accurate, computers are automatable devices that can perform a task without any user intervention.
- ✓ The user just needs to assign the task to the computer, after which it automatically controls different devices attached to it and executes the program instructions.

4. Diligence

- ✓ Computers never get tired of a repetitive task. It can continually work for hours without creating errors.
- ✓ Even if a large number of executions need to be executed, each and every execution requires the same duration, and is executed with the same accuracy.

5. Versatile

- ✓ Versatility is the quality of being flexible. Today, computers are used in our daily life in different fields.
- ✓ For example, they are used as personal computers (PCs) for home use, for business-oriented tasks, weather forecasting, space exploration, teaching, railways, banking, medicine, and so on, indicating that computers can perform different tasks simultaneously.

6. Memory

- ✓ Computers have internal or primary memory (storage space) as well as external or secondary memory.
- ✓ The computer stores a large amount of data and programs in the secondary storage space.
- ✓ The stored data and programs can be retrieved and used whenever required.
- ✓ Secondary memory is the key for data storage. Some examples of secondary devices include floppy disks, optical disks (CDs and DVDs), hard disk drives (HDDs), and pen drives.

7. No IQ

- ✓ Although the trend today is to make computers intelligent by inducing artificial intelligence (AI) in them, they still do not have any decision-making abilities of their own. They need guidance to perform various tasks.

8. Economical

- ✓ Today, computers are considered as short-term investments for achieving long-term gains. Using computers also reduces manpower requirements and leads to an elegant and efficient way of performing various tasks.
- ✓ Hence, computers save time, energy, and money. When compared to other systems, computers can do more work in lesser time.
- ✓ For example, using the conventional postal system to send an important document takes at least two to three days, whereas the same information when sent using the Internet (e-mail) will be delivered instantaneously.

1.3 Stored Program Concept

- ✓ All digital computers are based on the principle of stored program concept.
- ✓ The following are the key characteristic features of this concept:
 - Before any data is processed, **instructions are read into memory**.
 - **Instructions are stored in the computer's memory** for execution.
 - **Instructions are stored in binary form** (using binary numbers—only 0s and 1s).
 - **Processing starts with the first instruction** in the program, which is copied into a control unit circuit. The control unit executes the instructions.
 - **Instructions written by the users are performed sequentially until there is a break in the current flow.**
 - **Input/Output** and processing operations are performed simultaneously. While data is being read/written, the central processing unit (CPU) executes another program in the memory that is ready for execution.

1.3.1 Types of Stored Program Computers

1. Shared Memory Architecture

- ✓ A computer with a Von Neumann architecture stores data and instructions in the same memory.
- ✓ There is a serial machine in which data and instructions are selected one at a time.
- ✓ Data and instructions are transferred to and from memory through a shared data bus. Since there is a single bus to carry data and instructions, process execution becomes slower.

2. Distributed Memory Architecture

- ✓ Later Harvard University proposed a stored program concept in which there was a separate memory to store data and instructions.
- ✓ Instructions are selected serially from the instruction memory and executed in the processor. When an instruction needs data, it is selected from the data memory. Since there are separate memories, execution becomes faster.

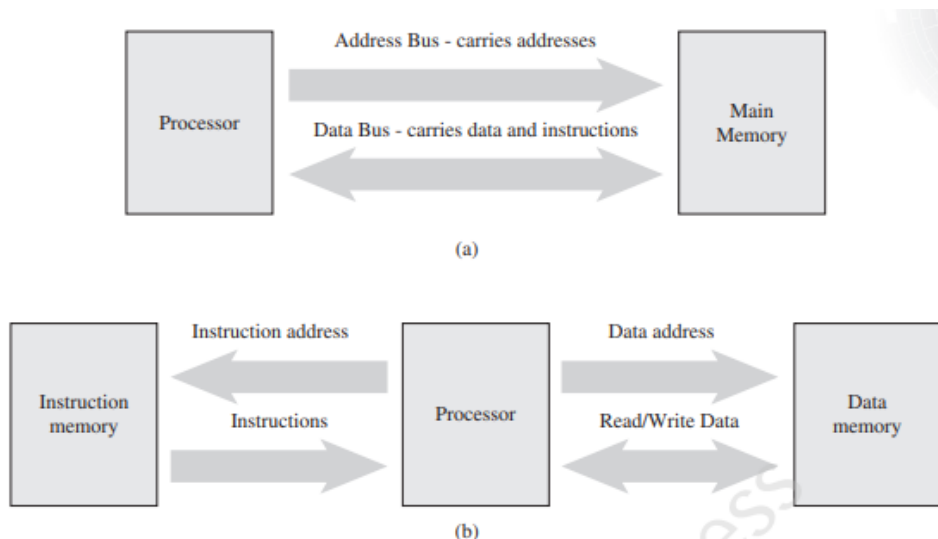


Figure 1.2. Von Neumann architecture (a) Shared memory for instructions and data (b) Separate memories for instructions and data

1.4 History of Computers

✓ Timeline of Developments

300 BC: The abacus was an early aid for mathematical computations and was designed to aid human's memory while performing calculations.

1822: Charles Babbage designed a steam-driven calculating machine that could compute tables of numbers.

1890: a punched card system was designed for the calculations.

1936: Turing machine were designed, which were capable of computing anything that is computable.

1941: It was the first time a computer could store information in its main memory.

1943–1944:

- Electronic Numerical Integrator and Calculator (ENIAC) were built.
- It is considered as the grandfather of digital computers.
- It filled a 20×40 feet room and had 18,000 vacuum tubes.

1946: UNIVAC was designed, which was the first commercial computer for business and government applications.

1947: vacuum tubes in computers were replaced by transistors.

1953: Developed the first computer language COBOL.

1954: The FORTRAN programming language was developed.

1958: invented integrated circuit, which is commonly known as the computer chip.

1964: developed a prototype of the modern computer, with a mouse and a graphical user interface (GUI). This was a remarkable achievement as it shifted computers from a specialized machine for scientists and mathematicians to general public.

1969: Unix operating system was developed at Bell Labs. It was written in the C programming language and was designed to be portable across multiple platforms.

1970: DRAM chip was introduced by Intel.

1971: invented the floppy disk which allowed data to be shared among computers.

1973: developed Ethernet for connecting multiple computers and other hardware.

1974–1977: Personal computers started becoming popular.

1975: Paul Allen and Bill Gates started writing software using the new BASIC language. They both formed their own software company, Microsoft.

1976: Apple Computers and developed Apple I, the first computer with a single-circuit board.

1977: Apple II was launched that offered colour graphics and incorporated an audio cassette drive for storage.

1978: a word processor application, was released by MicroPro International.

1979: the first computerized spreadsheet program for personal computers, was unveiled.

1981: The first IBM personal computer was introduced that used Microsoft's MS-DOS operating system. The term PC was popularized.

1983: The first laptop was introduced.

1985: Microsoft announced Windows as a new operating system.

1986: Compaq introduced Deskpro 386 in the market, which was a 32-bit architecture machine that provides speed comparable to mainframes.

1990: Invented the World Wide Web with HTML as its publishing language.

1993: The Pentium microprocessor introduced the use of graphics and music on PCs.

1994: PC games became popular.

1996: developed the Google search engine at Stanford University.

1999: The term Wi-Fi was introduced when users started connecting to the Internet without wires.

2001: Apple introduced Mac OS X operating system, which had protected memory architecture and pre-emptive multi-tasking, among other benefits. To stay competitive, Microsoft launched Windows XP.

2003: The first 64-bit processor, AMD's Athlon 64, was brought into the consumer market.

2004: Mozilla released Firefox 1.0 and in the same year Facebook, a social networking site, was launched.

2005: YouTube, a video sharing service, was launched. In the same year, Google acquired Android, a Linux-based mobile phone operating system.

2006: Apple introduced MacBook Pro, its first Intelbased, dual-core mobile computer.

2007: Apple released iPhone, which brought many computer functions in the smartphone.

2009: Microsoft launched Windows 7 in which users could pin applications to the taskbar.

2010: Apple launched iPad, which revived the tablet computer segment.

2011: Google introduced Chromebook, a laptop that runs on the Google Chrome operating system.

2015: Apple released the Apple Watch. In the same year, Microsoft launched Windows 10.

1.4.1 Generations of Computers

1. First Generation (1942–1955)

Hardware Technology

First generation computers were manufactured using thousands of ***vacuum tubes***.

Memory

- **Electromagnetic** relay was used as **primary memory** and
- **Punched cards** were used to **store data and instructions**.

Software Technology

- Programming was done in machine or assembly language.

Used for Scientific applications

Examples ENIAC, EDVAC, EDSAC, UNIVAC I, IBM 701

Highlights

- They were the fastest calculating device.
- Computers were too bulky and required a complete room for storage
- Highly unreliable as vacuum tubes emitted a large amount of heat and burnt frequently

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- Required air-conditioned rooms for installation
 - Costly
 - Difficult to use
 - Required constant maintenance because vacuum tubes used filaments that had limited life time. Therefore, these computers were prone to frequent hardware failures.

2. Second Generation (1955–1964)

Hardware Technology

- Second generation computers were manufactured using **transistors**.
- Transistors were reliable, powerful, cheaper, smaller, and cooler than vacuum tubes.

Memory

- Magnetic core memory was used as primary memory;
- Magnetic tapes and magnetic disks were used to store data and instructions.
- These computers had faster and larger memory than the first generation computers.

Software Technology

- Programming was done in high level programming languages. Batch operating system was used.

Used for Scientific and commercial applications

Examples Honeywell 400, IBM 7030, CDC 1604, UNIVAC LARC

Highlights

- Faster, smaller, cheaper, reliable, and easier to use
- They consumed 1/10th the power consumed by first generation computers
- Bulky in size
- Dissipated less heat than first generation computers
- Costly
- Difficult to use.

3. Third Generation (1964–1975)

Hardware Technology

- Third generation computers were manufactured using integrated chips (ICs).
- ICs consist of several components such as transistors, capacitors, and resistors on a single chip to avoid wired interconnections between components.
- These computers used SSI and MSI technology. Minicomputers came into existence.

Memory

- Larger magnetic core memory was used as primary memory; larger capacity magnetic tapes and magnetic disks were used to store data and instructions.

Used for Scientific, commercial, and interactive online applications.

Highlights

- Faster, smaller, cheaper, reliable, and easier to use than the second generation computers
- They consumed less power than second generation computers.
- Bulky in size and required a complete room for installation
- Dissipated less heat than second generation computers
- Costly

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- Easier to use and upgrade

Software Technology

- Programming was done in high level programming languages such as FORTRAN, COBOL, Pascal, and BASIC.
- Time sharing operating system was used.
- Software was separated from the hardware.

4. Fourth Generation (1975–1989)

Hardware Technology

- Fourth generation computers were manufactured using ICs with LSI (Large Scale Integrated) and later with VLSI technology (Very Large Scale Integration).
- Microcomputers came into existence.
- Use of personal computers became widespread.
- High speed computer networks in the form of LANs, WANs, and MANs started growing.

Memory

- Semiconductor memory was used as primary memory; large capacity magnetic disks were used as built in secondary memory.
- Magnetic tapes and floppy disks were used as portable storage devices.

Software Technology

- Programming was done in **high level programming language such as C and C++**.
- **Graphical User Interface** (GUI) based operating system (e.g. Windows) was introduced. It had **icons and menus** among other features to allow computers to be used as a general purpose machine by all users.
- UNIX was also introduced as an open source operating system. Apple Mac OS and MS DOS were also released during this period.
- All these operating systems had multi-processing and multiprogramming capabilities.

Used for Scientific, commercial, interactive online, and network applications.

Examples IBM PC, Apple II, TRS-80, VAX 9000, CRAY1, CRAY-2, CRAY-X

Highlights

- Faster, smaller, cheaper, powerful, reliable, and easier to use than the previous generation computers.

5. Fifth Generation (1989–Present)

Hardware Technology

- Fifth generation computers are manufactured using ICs with ULSI (Ultra Large Scale Integrated) technology.
- The use of Internet became widespread and very powerful mainframes, desktops, portable laptops, and smartphones are being used commonly.
- Supercomputers use parallel processing techniques.

Memory

- Semiconductor memory is used as primary memory;
- large capacity magnetic disks are used as built-in secondary memory.
- portable storage devices like optical disks and USB flash drives are used.

Software Technology

- Programming is done in high-level programming languages such as Java, Python, and C#.
- Graphical User Interface (GUI)-based operating systems such as Windows, Unix, Linux, Ubuntu, and Apple Mac are being used.
- These operating systems are more powerful and user friendly than the ones available in the previous generations.

Used for Scientific, commercial, interactive online, multimedia (graphics, audio, video), and network applications.

Examples IBM notebooks, Pentium PCs, SUM workstations, IBM SP/2, Param supercomputer

Highlights

- Faster, smaller, cheaper, powerful, reliable, and easier to use than the previous generation computers.
- Speed of microprocessors and the size of memory are growing rapidly.
- High-end features available on the microprocessors
- They consume less power than computers of prior generations.

1.5 Classification of Computers

✓ Computers can be broadly classified into four categories based on their speed, amount of data that they can process, and price:

- Supercomputers
- Mainframe computers
- Minicomputers
- Microcomputers

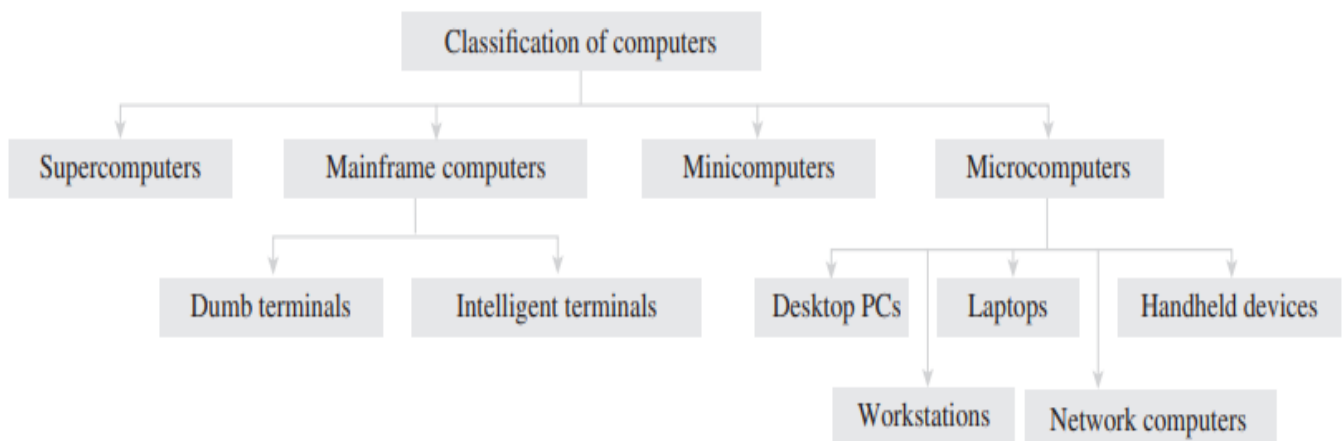


Figure 1.3. Classification of computers

1. Supercomputers

- ✓ The supercomputer is the fastest, most powerful, and most expensive computer.
- ✓ Process large amounts of data and to solve complex scientific problems.
- ✓ Supercomputers use parallel processing technology and can perform more than one trillion calculations in a second.
- ✓ A single supercomputer can support thousands of users at the same time.
- ✓ Such computers are mainly used for weather forecasting, nuclear energy research, aircraft design, automotive design, online banking, controlling industrial units, etc.

2. Mainframe Computers

- ✓ Mainframe computers are large-scale computers (but smaller than supercomputers).
- ✓ These are very expensive and need a very large clean room with air conditioning, thereby making them very costly to deploy.
- ✓ Support multiple processors.
- ✓ The two types of terminals that can be used with mainframe systems are as follows:

1. Dumb Terminals

- ✓ Dumb terminals consist of only a monitor and a keyboard (or mouse). They do not have their own CPU and memory and use the mainframe system's CPU and storage devices.

2. Intelligent Terminals:

- ✓ Intelligent terminals have their own processor and thus can perform some processing operations.
- ✓ Usually, PCs are used as intelligent terminals to facilitate data access and other services from the mainframe system.
- ✓ Mainframe computers are typically used as servers on the World Wide Web.
- ✓ They are also used in organizations such as banks, airline companies, and universities, where a large number of users frequently access the data stored in their databases.

3. Minicomputers

- ✓ Minicomputers are smaller, cheaper, and slower than mainframes.
- ✓ They are called minicomputers because they were the smallest computer of their times. Also known as midrange computers.
- ✓ Minicomputers are widely used in business, education, hospitals, government organizations, etc. While some minicomputers can be used only by a single user, others are specifically designed to handle multiple users simultaneously.
- ✓ Minicomputers can also be used as servers in a networked environment.

4. Microcomputers

- ✓ Microcomputers, commonly known as PCs, are very small and cheap.
- ✓ Many computer hardware companies copied this design and termed their microcomputers as PC-compatible.
- ✓ PCs can be classified into the following categories:

Desktop PCs, Laptops, Workstations, Network Computers and Handheld Computers.

(i) Desktop PCs A desktop PC is the most popular model of PCs.

(ii) Laptops: Laptops operate on a battery. The memory and storage capacity of a laptop is almost equivalent to that of a desktop computer. laptops have the same features and processing speed as the most powerful PCs.

(iii) Workstations: Workstation computers have advanced processors, more RAM and storage capacity than PCs.

(iv) Network Computers: Network computers have less processing power, memory, and storage than a desktop computer. Some network computers do not have any storage space and merely rely on the network's server for data storage and processing tasks

(v) Handheld Computers: Handheld computers are very small in size, and hence they have small-sized screens and keyboards. Some examples of handheld computers are as follows: Smartphones, Tablet PCs.

a. Smartphones: These days, cellular phones are web-enabled telephones. Such phones are also known as smartphones because, in addition to basic phone capabilities, they also facilitate the users to access the Internet and send e-mails, edit Word documents, generate an Excel sheet, create a presentation, and lots more.

b. Tablet PCs: A tablet PC is a computing device that is smaller than a laptop, but bigger than a smartphone. Features such as user-friendly interface, portability, and touch screen have made them very popular in the last few years.

Uses The following are the uses of Tablet PCs:

- View presentations
- Videoconferencing
- Reading e-books, e-newspaper
- Watching movies
- Playing games
- Sharing pictures, video, songs, documents, etc.
- Browsing the Internet
- Keeping in touch with friends and family on popular social networks, sending emails
- Business people use them to perform tasks such as editing a document, exchanging documents, taking notes, and giving presentations
- Tablets are best used in crowded places such as airports and coffee shops.

1.6 Applications of Computers

1. Word processing

- ✓ Word processing software enables users to read and write documents.
- ✓ Users can also add images, tables, and graphs for illustrating a concept.
- ✓ The software automatically corrects spelling mistakes and includes copy–paste features.

2. Internet

- ✓ The Internet is a network of networks that connects computers all over the world.
- ✓ It gives the user access to an enormous amount of information, much more than available in any library.
- ✓ Using e-mail, the user can communicate in seconds with a person who is located thousands of miles away.
- ✓ Chat software enables users to chat with another person in real-time.
- ✓ Video conferencing tools are becoming popular.

3. Digital video or audio composition

- ✓ Computers make audio or video composition and editing very simple.
- ✓ Graphics engineers use computers for developing short or full-length films and creating 3-D models and special effects in science fiction and action movies.

4. Desktop publishing

- ✓ Desktop publishing software enables us to create page layouts for entire books.

5. e-Business

- ✓ e-Business or electronic business is the process of conducting business via the Internet.
- ✓ This may include buying and selling of goods and services using computers and the Internet.
- ✓ Companies today use e-commerce applications for marketing, transaction, processing, and product and customer services processing.

Business-to-consumer or B2C: In this form of electronic commerce, business companies deploy their websites on the Internet to sell their products and services to the customers. On their websites, they provide features such as catalogues, interactive order processing system, secure electronic payment system, and online customer support.

Business-to-business or B2B: This type of electronic commerce involves business transactions performed between business partners (customers are not involved). For example, companies use computers and networks (in the form of extranets) to order raw materials from their suppliers. Companies can also use extranets to supply their products to their dealers.

Consumer-to-consumer or C2C: This type of electronic commerce enables customers to carry business transactions among themselves. For example, on auction websites, a customer sells his/her product which is purchased by another customer.

Electronic banking **Electronic banking**, also known as cyber banking or online banking, supports various banking activities conducted from home, a business, or on the road instead of a physical bank location.

6. Bioinformatics

- ✓ Bioinformatics is the application of computer technology to manage large amount of biological information.
- ✓ Computers are used to collect, store, analyse, and integrate biological and genetic information to facilitate gene-based drug discovery and development.
- ✓ Bioinformatics is an interdisciplinary field of molecular biology, computer science, statistics, and mathematics.

7. Health care

- ✓ Computers have also become a necessary device in the health care industry. The following are areas in which computers are extensively used in the health care industry:
Storing records: computers are first and foremost used to store the medical records of patients.
Surgical procedures Computers are used for certain surgical procedures. They enable the surgeon to use computer to control and move surgical instruments in the patient's body for a variety of surgical procedures.
Better diagnosis and treatment Computers help physicians make better diagnoses and recommend treatments.

8. Geographic Information System and Remote Sensing

- ✓ A geographic information system (GIS) is a computer based tool for mapping and analysing earth.
- ✓ It integrates database operations and statistical analysis to be used with maps.
- ✓ Remote sensing is a sub-field of geography, which can be applied in the following areas to collect data of dangerous or inaccessible areas for the following: Monitoring deforestation, Studying features of glaciers, etc.

9. Meteorology

- ✓ Meteorology is the study of the atmosphere. Meteorology has applications in many diverse fields. Some of the applications include the following:
- ✓ Weather forecasting It includes application of science and technology to predict the state of the atmosphere (temperature, precipitation, etc.) for a future time and a given location.
- ✓ Aviation meteorology Aviation meteorology studies the impact of weather on air traffic management.
- ✓ Agricultural meteorology Agricultural meteorology deals with the study of effects of weather and climate on plant distribution, crop yield, water-use efficiency, plant and animal development.
- ✓ Nuclear meteorology Nuclear meteorology studies the distribution of radioactive aerosols and gases in the atmosphere. Maritime meteorology Maritime meteorology is the study of air and wave forecasts for ships operating at sea.
- ✓ Multimedia and Animation Multimedia and animation that combines still images, moving images, text, and sound in meaningful ways is one of most powerful aspects of computer technology. We all have seen cartoon movies, which are nothing but an example of computer animation.

10. Legal System

- ✓ Lawyers use computers to look through millions of individual cases and find whether similar or parallel cases have been approved, denied, criticized, or overruled in the past.

11. Retail Business

- ✓ Computers are used in retail shops to enter orders, calculate costs, and print receipts.

12. Sports

- ✓ In sports, computers are used to compile statistics, identify weak players and strong players by analysing statistics, sell tickets, create training programs and diets for athletes, and suggest game plan strategies based on the competitor's past performance.

13. Travel and Tourism

- ✓ Computers are used to prepare tickets, monitor the train's or airplane's route, and guide the plane to a safe landing.

14. Simulation

- ✓ Simulation of automobile crashes or airplane emergency landings is done to identify potential weaknesses in designs without risking human lives.

15. Education

- ✓ A computer is a powerful teaching aid and can act as another teacher in the classroom. Teachers use computers to develop instructional material.

16. Industry and Engineering

- ✓ Computers are found in all kinds of industries, such as thermal power plants, oil refineries, and chemical industries, for process control, computer-aided designing (CAD), and computer-aided manufacturing (CAM).

17. Robotics

- ✓ Robots are computer-controlled machines mainly used in the manufacturing process in extreme conditions where humans cannot work.

18. Decision Support Systems

- ✓ Computers help managers to analyse their organization's data to understand the present scenario of their business, view the trends in the market, and predict the future of their products.

19. Expert Systems

- ✓ Expert systems are used to automate the decision-making process in a specific area, such as analysing the credit histories for loan approval and diagnosing a patient's condition for prescribing an appropriate treatment.

1.7 Basic Organization of a Computer

- ✓ A computer is an electronic device that performs five major operations:
 - Accepting data or instructions (input)
 - Storing data
 - Processing data
 - Displaying results (output)
 - Controlling and coordinating all operations inside a computer.

Input: This is the process of entering data and instructions (also known as programs) into the computer system. The data and instructions can be entered by using different input devices such as keyboard, mouse, scanner, and trackball.

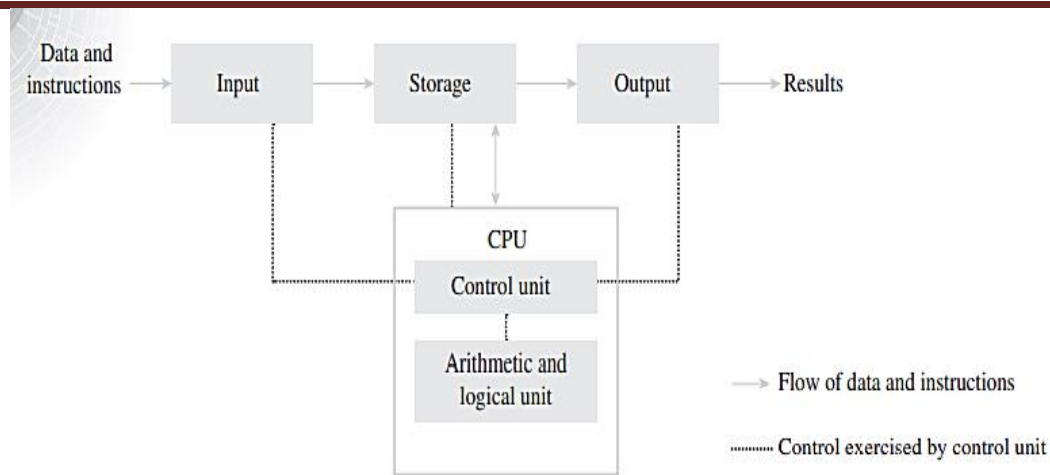


Figure 1.4. Block diagram of a computer

Storage: Storage is the process of saving data and instructions permanently in the computer so that they can be used for processing.

A computer has two types of storage areas:

Primary storage: Primary storage, also known as the main memory, is the storage area that is directly accessible by the CPU at very high speeds. Primary storage space is very expensive and therefore limited in capacity. Another drawback of main memory is that it is volatile in nature; that is, as soon as the computer is switched off, the information stored gets erased. Hence, it cannot be used as a permanent storage of useful data and programs for future use. Example of primary storage is random access memory (RAM).

Secondary storage Also known as auxiliary memory. It is cheaper, non-volatile, and used to permanently store data and programs of those jobs that are not being currently executed by the CPU.

Output: Output is the process of giving the result of data processing to the outside world. The results are given through output devices such as monitor, and printer. Since the computer accepts data only in binary form and the result of processing is also in binary form, the result cannot be directly given to the user. The output devices, therefore, convert the results available in binary codes into a human-readable language before displaying it to the user.

Control: The control unit (CU) manages and controls all the components of the computer system. It is the CU that decides the manner in which instructions will be executed and operations performed. It takes care of the step-by-step processing of all operations that are performed in the computer.

The CPU is a combination of the **arithmetic logic unit (ALU)** and the **CU**. The CPU is better known as the brain of the computer system because the entire processing of data is done in the ALU, and the CU activates and monitors the operations of other units (such as input, output, and storage) of the computer system.

Processing: The process of performing operations on the data as per the instructions specified by the user (program) is called processing. Data and instructions are taken from the primary memory and transferred to the ALU, which performs all sorts of calculations.

1.8 Motherboard

- ✓ The motherboard, also known as the main board or the parent board is the primary component of a computer. It is used to connect all the components of the computer.

Characteristics of a Motherboard

✓ A motherboard can be classified depending on the following characteristics:

- Form factor
- Chipset
- Type of processor socket used
- Input–Output connectors

Form factor: Form factor refers to the motherboard's geometry, dimensions, arrangement, and electrical requirements.

Integrated components: Some of the motherboard's components are integrated into its printed circuitry. These include the following:

- The chipset is a circuit that controls the majority of the computer's resources such as the bus interface with the processor, cache memory, RAM, and expansion cards.
- CMOS clock and battery
- BIOS
- System bus and expansion bus

Chipset: The chipset is an electronic circuit that basically coordinates data transfers between the different components of the computer.

Some chipsets may include a graphics or audio chip, which makes it unnecessary to install a separate graphics card or sound card.

CMOS clock and battery: The real-time clock (or RTC) is a circuit that is used to synchronize the computer's signals. When the computer is switched off, the power supply stops providing electricity to the motherboard. The CMOS chip is powered by a battery located on the motherboard.

Processor socket: The processor (also called the microprocessor) is the brain of the computer. The processor is characterized by its speed or frequency, which is the rate at which it executes instructions.

RAM connectors RAM is the primary storage area that stores data while the computer is running. However, its contents are erased when the computer is turned off or restarted.

Expansion slots: Expansion slots are compartments into which expansion cards can be inserted. Such cards render new features or enhance the computer's performance.

I/O connectors: The motherboard has a number of input– output sockets on its rear panel, some of which include:

- A serial port to connect some old peripherals
- A parallel port to connect old printers
- USB ports to connect more recent peripherals such as mouse and pen drive.
- RJ45 connector (also known as LAN or Ethernet port) to connect the computer to a network. It corresponds to a network card integrated into the motherboard.
- Video graphics array (VGA) connector to connect a monitor. This connector interfaces with the built-in graphics card.
- Audio plugs that include the line in, line out, and microphone to connect sound speakers, hi-fi system, or microphone. This connector interfaces with the built-in sound card.

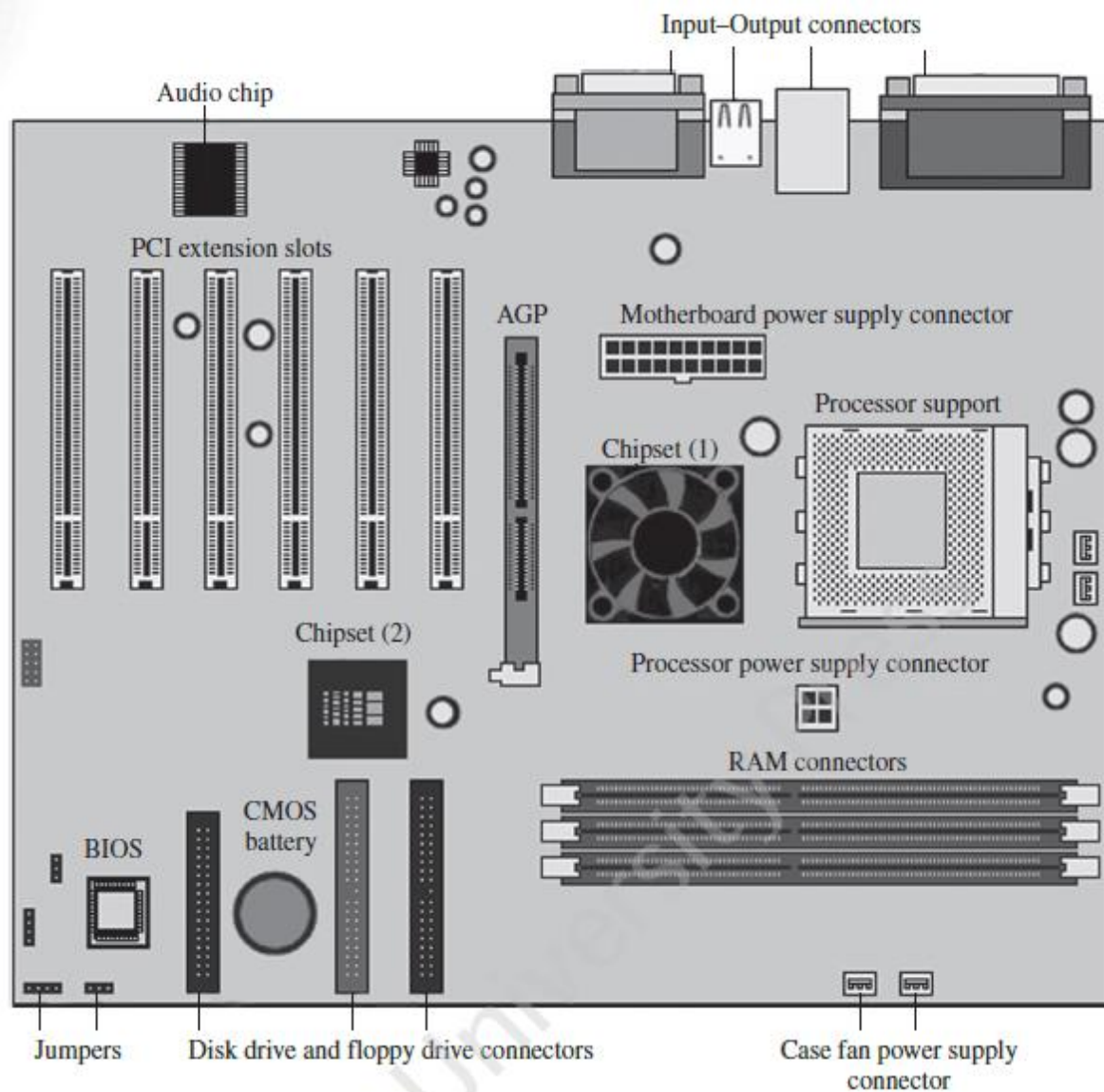


Figure 1.5. Computer's motherboard